박범수

A. 교수님과 연구실 소개

교수님 약력

- **서울과학기술대학교 MSDE학과 조교수**
- KAIST 기계기술연구소 박사후연구원 (23′~26′)
- Rensselaer Polytechnic Institute 기계공학 박사 (23′)
- 울산과학기술원 기계공학 석사 (19′)
- 울산과학기술원 기계항학 학사 (17′)

연구실 정보

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- 전화: 02-970-7298 (010-4040-8929)
- 이메일: bspark@seoultech.ac.kr
- 연구실명: 제조인공지능 연구실 (MAI Lab)
- Google Scholar: *https://scholar.google.com/citations?user=-8DQcmYAAAAJ&hl=en&oi=ao*
- 연구실 홈페이지: *https://mai.seoultech.ac.kr*



대표 연구 키워드

Physics-informed AI

- 물리 법칙을 결합한 기계 시스템의 고속·고신뢰 해석 모델 개발

AI-enabled Metrology

- 딥러닝 기반 3D 유동장 계측과 제조 시스템 고장 검출·진단

Generative Inverse Design

- 대리 모델 기반 공정 최적화 및 생성형 AI 역설계

AI Control / Physical AI

- Sim-to-real 강화학습 기반 공정 제어 및 자율 제조

기계 인공 지능 연구회

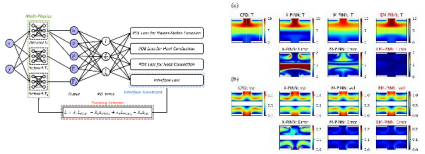
산학 협력 탐색 그룹

박범수

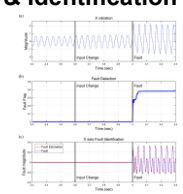
B. 주요 연구 분야 (연구실 개요)

Analysis, Metrology, & Diagnostics

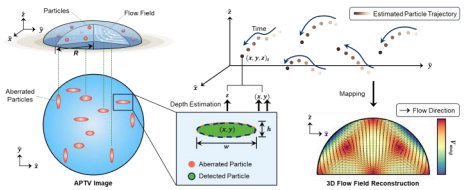
Physics-informed ML

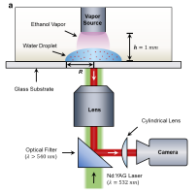


Fault Detection & Identification

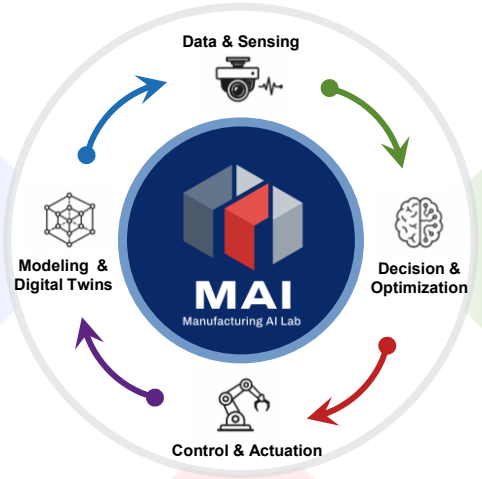


AI-enabled Metrology



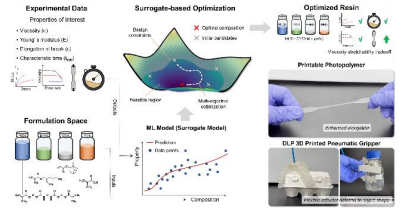


AI for Understanding Manufacturing Systems

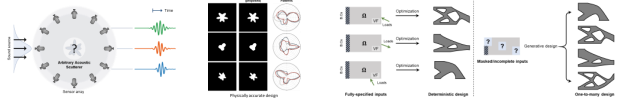


Inverse Design & Process Optimization

Surrogate-based Process Parameter Optimization



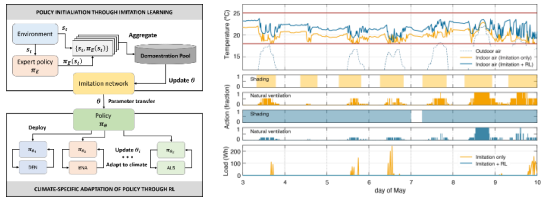
Generative Inverse Design



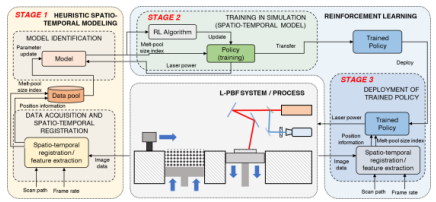
AI for Designing Better Processes and Materials

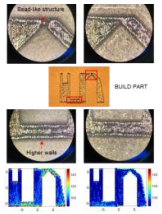
AI-based Control & Physical AI

Reinforcement Learning Control / Imitation Learning



Sim-to-Real Learning for Process Control





Toward Physical AI / Autonomous Manufacturing

- Autonomous Decision Making
- Self-optimization & Adaptation
- Closed-loop Autonomous Systems

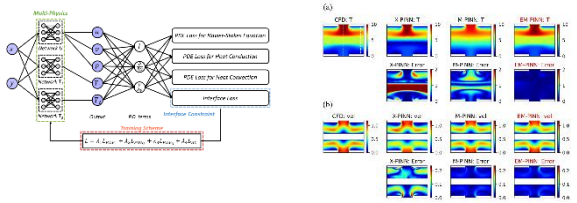
AI for Autonomous and Adaptive Manufacturing

박범수

B. 주요 연구성과 (논문 중심)

Physics-informed Digital Twins

Physics-informed ML

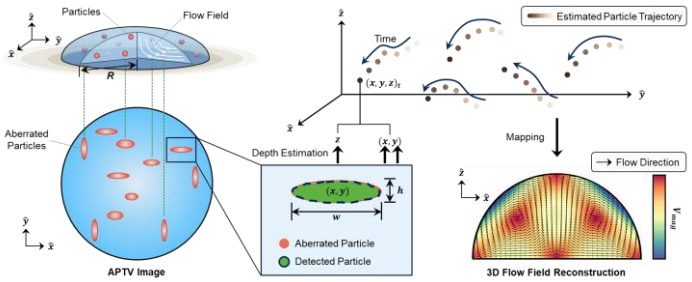


International Journal of Heat and Mass Transfer, 2025

Scientific Reports, 2025

Engineering Applications of Artificial Intelligence, 2026

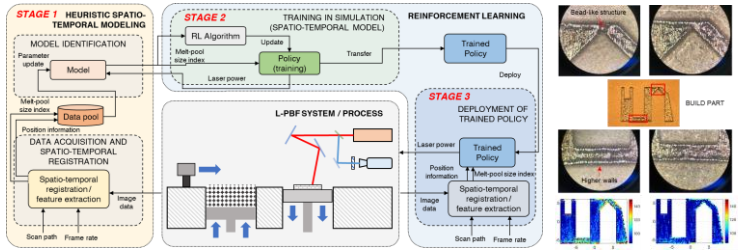
AI-enabled metrology



Measurement, 2026

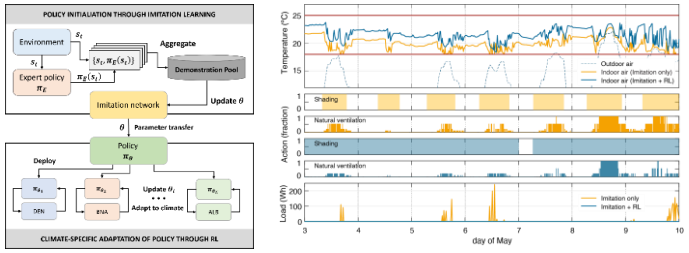
Process Control / Physical AI

Sim-to-real Reinforcement Learning



IEEE TASE, 2024

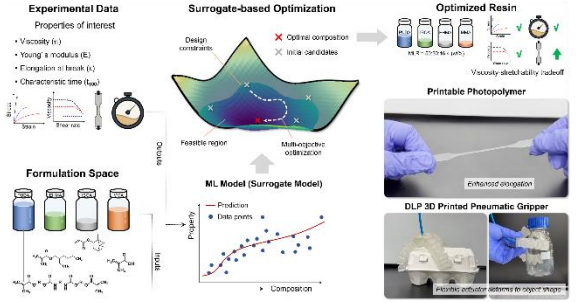
Expert-assisted Reinforcement Learning



Applied Energy, 2023

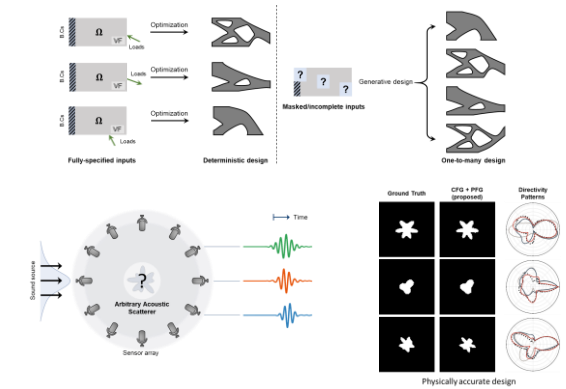
Process Optimization / Design

Surrogate-based Optimization



Nature Communications, 2026

Inverse design with Generative Models



기계 인공 지능 연구회

산학 협력 탐색 그룹